Gabor Vector-Field Model Comparison with Fixed Hyperparameter Sweeps

Rishabh Kumar

May 21, 2026

Abstract

This report compares parametric spiral families and non-parametric kernel/RKHS families on the Gabor-filter vector-field exports in `vf_exports`. Unlike the earlier adaptive-grid analysis, every kernel model here uses the same broad fixed hyperparameter grids. The primary score is held-out axial residual sum of squares (RSS), not Bayesian evidence. The plots overlay fitted streamlines on the original Gabor arrows, with the observed field drawn at opacity $\alpha = 0.7$.

1 Data and Angular Representation

Each Gabor CSV row supplies a coordinate (x_i, y_i) and the local pitch angle α'_i in degrees. Because α'_i is not the global field direction, the observed modelling angle is reconstructed as

$$\phi_i = \operatorname{atan2}(y_i, x_i) + \alpha'_i.$$

The models are scored as line fields, so ϕ and $\phi + \pi$ are equivalent. Held-out error uses the axial residual

$$\Delta_i = \frac{1}{2}\operatorname{atan2}\left(\sin 2(\phi_i - \hat{\phi}_i), \cos 2(\phi_i - \hat{\phi}_i)\right).$$

The reported RSS is the mean across shuffled 10-fold cross-validation folds of the per-held-out-point sum $\sum_i \Delta_i^2/|I_k|$.

2 Fixed Hyperparameter Sweep

The kernel sweep is intentionally wide and fixed across datasets. It does not use the dataset-adaptive length-scale or noise ranges from the earlier analysis.

Hyperparameter	Count	Min	Max	Grid
bandwidth_or_radius_ell	72	1	1.5e+03	geomspace
sigma_n	17	0.001	10	geomspace
kappa	19	0	64	explicit
parametric_fixed_p	3	0	2	{0,1,2}
parametric_continuous_p_bounds		-0.999	2.999	L-BFGS-B bounds
parametric_gamma_bounds		-3.141593	3.141593	L-BFGS-B bounds

The Gaussian RBF and uniform kernels each evaluate $72 \times 17 = 1224$ kernel–noise settings per dataset. The multiplicative RBF–von-Mises kernel evaluates $72 \times 17 \times 19 = 23256$ settings per dataset. Each setting is scored by shuffled 10-fold held-out axial RSS with random seed 42.

3 Model Families

The non-parametric families are Gaussian RBF, uniform local-neighbourhood, and multiplicative RBF–von-Mises kernels on the doubled-angle embedding $(\cos 2\phi, \sin 2\phi)$. The parametric families are fixed $p \in \{0, 1, 2\}$ and a continuous p family with $p \in [-0.999, 2.999]$ and $\gamma \in [-\pi, \pi]$.

4 Results

4.1 Winner Counts

Model	Wins
Multiplicative RBF-VM RKHS	6
Gaussian RBF RKHS	2
Uniform RKHS	2

4.2 Held-Out RSS Summary

Model	Datasets	Mean RSS	Median RSS	Mean MAE deg
Multiplicative RBF-VM RKHS	10	0.3396	0.3358	24.7723
Gaussian RBF RKHS	10	0.342	0.3417	24.8753
Uniform RKHS	10	0.3576	0.3488	25.8737
Parametric p=0 (Logarithmic)	10	0.7851	0.78	43.843
Parametric p=1 (Archimedean)	10	0.7986	0.7736	43.964
Parametric continuous p	10	0.8045	0.8138	44.3255
Parametric p=2 (Fermat)	10	0.8278	0.8407	44.8518

4.3 Selected Kernel Hyperparameters

Model	ℓ min	ℓ med	ℓ max	σ_n min	σ_n med	σ_n max	κ min	κ med	κ max
Gaussian RBF RKHS	24.3637	50.3709	103.0423	0.001	0.5623	10			
Multiplicative RBF-VM RKHS	24.3637	52.8229	319.9529	0.0178	1	3.1623	0	0.625	3
Uniform RKHS	45.2008	58.5539	68.2466	0.0562	5.6234	10			

4.4 Dataset Winners

Dataset	Winner	RSS	MAE deg
Front_EE-1_1_3000x_rings_coords.csv	Multiplicative RBF-VM RKHS	0.2587	20.7635
Front_EE-1_2_3000x_rings_coords.csv	Multiplicative RBF-VM RKHS	0.2194	18.4802
Front_EE-1_3_3000x_rings_coords.csv	Gaussian RBF RKHS	0.4909	31.8122
Front_EE-1_4_3000x_rings_coords.csv	Gaussian RBF RKHS	0.388	27.623
Front_EE-1_5_3000x_rings_coords.csv	Multiplicative RBF-VM RKHS	0.34	24.4879
Front_EE1_1_3000x_rings_coords.csv	Multiplicative RBF-VM RKHS	0.15	16.0471
Front_EE1_2_3000x_rings_coords.csv	Uniform RKHS	0.4622	29.9347
Front_EE1_3_3000x_rings_coords.csv	Uniform RKHS	0.3312	25.0436
Front_EE1_4_3000x_rings_coords.csv	Multiplicative RBF-VM RKHS	0.2572	22.3987
Front_EE1_5_3000x_rings_coords.csv	Multiplicative RBF-VM RKHS	0.4834	30.8891

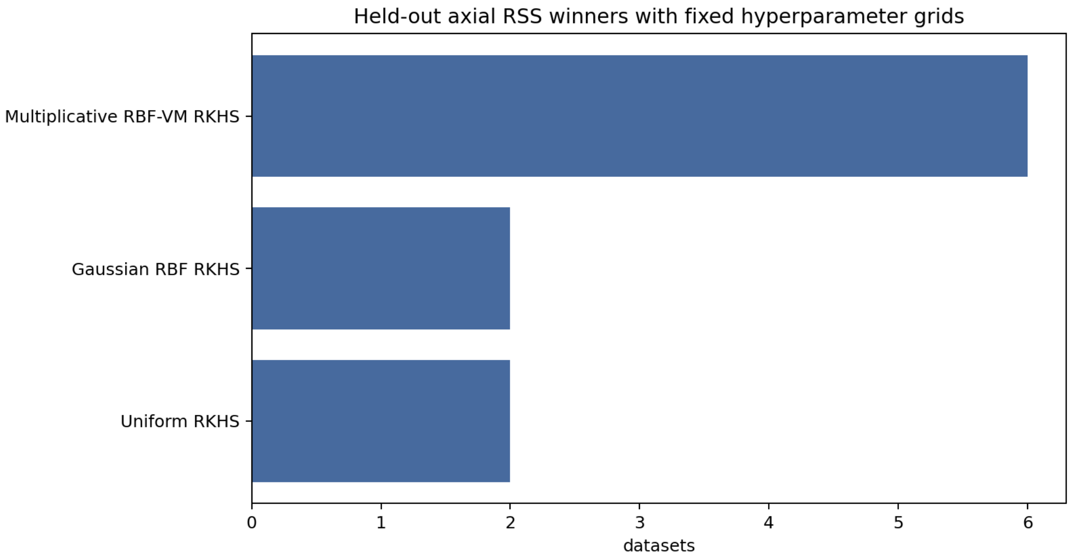

Figure 1: Winner counts under the fixed-grid held-out RSS comparison.

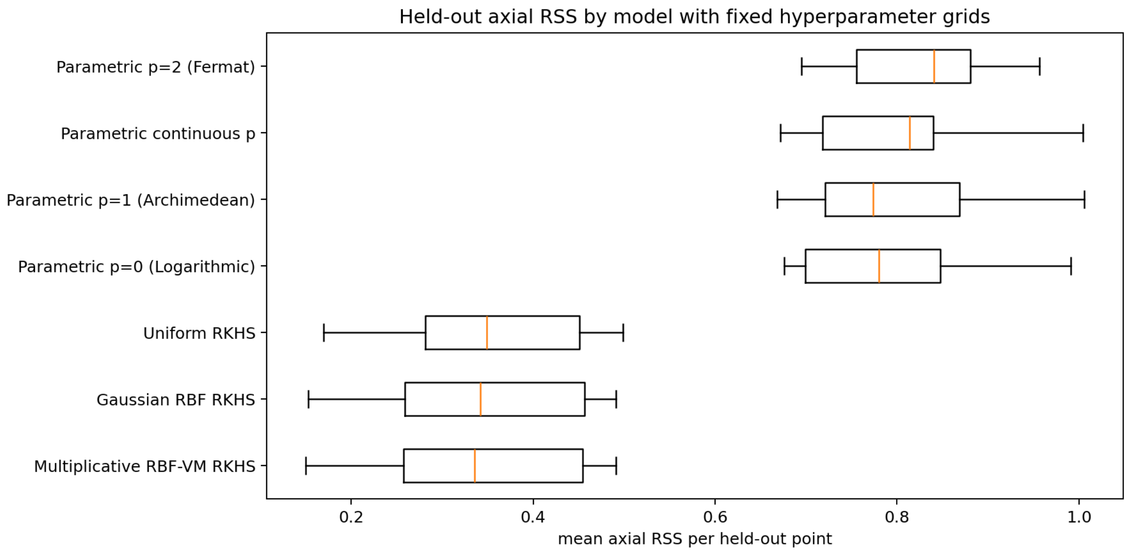

Figure 2: Distribution of held-out axial RSS across the 10 Gabor datasets.

5 Fitted Overlay Plots

Each plot shows the original Gabor vector field and the fitted streamlines for all compared families. The Gabor arrows come from α' after conversion to the global angle $\phi = \operatorname{atan2}(y, x) + \alpha'$.

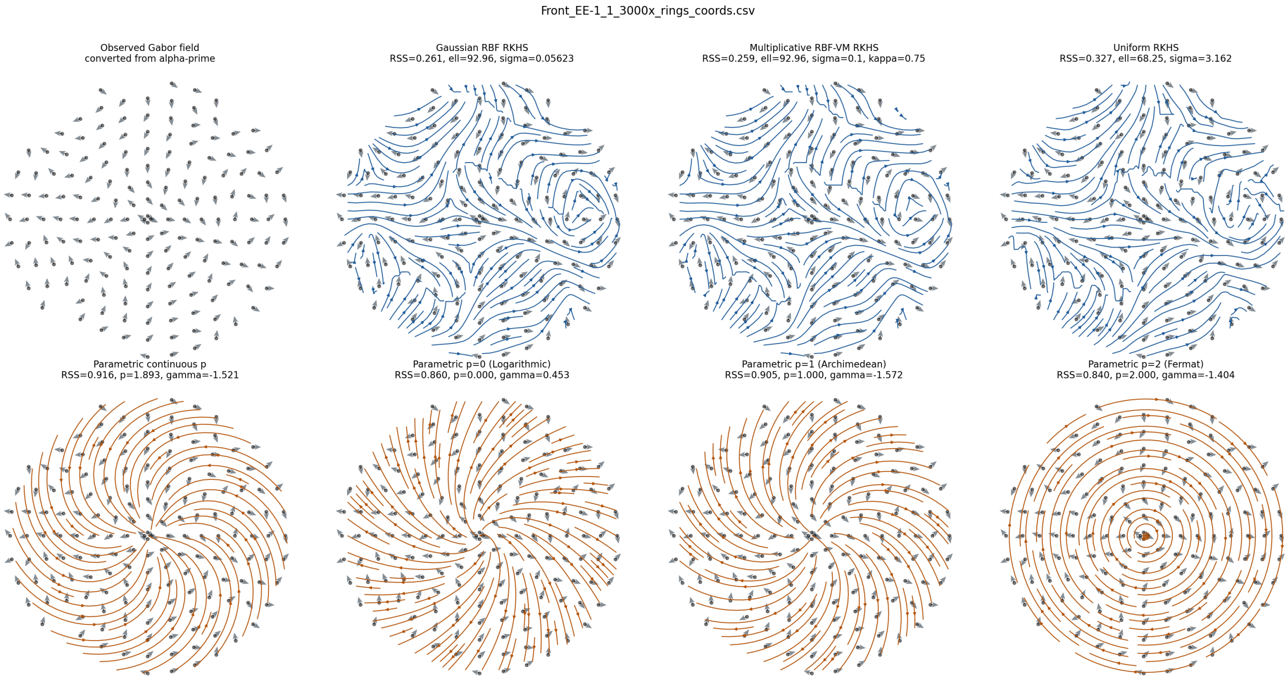

Figure 3: Fitted overlays for `Front_EE-1_1_3000x_rings_coords.csv`. The observed Gabor field is shown with arrow opacity $\alpha = 0.7$; fitted models are shown as streamlines.

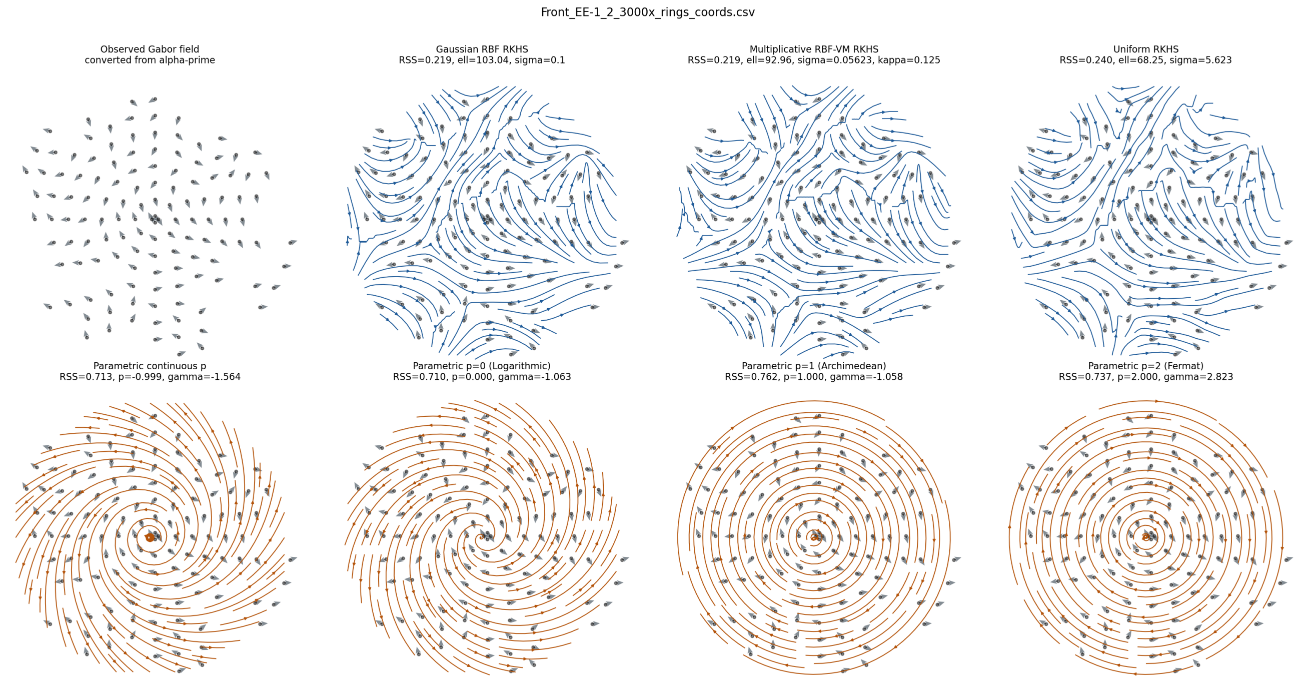

Figure 4: Fitted overlays for `Front_EE-1_2_3000x_rings_coords.csv`. The observed Gabor field is shown with arrow opacity $\alpha = 0.7$; fitted models are shown as streamlines.

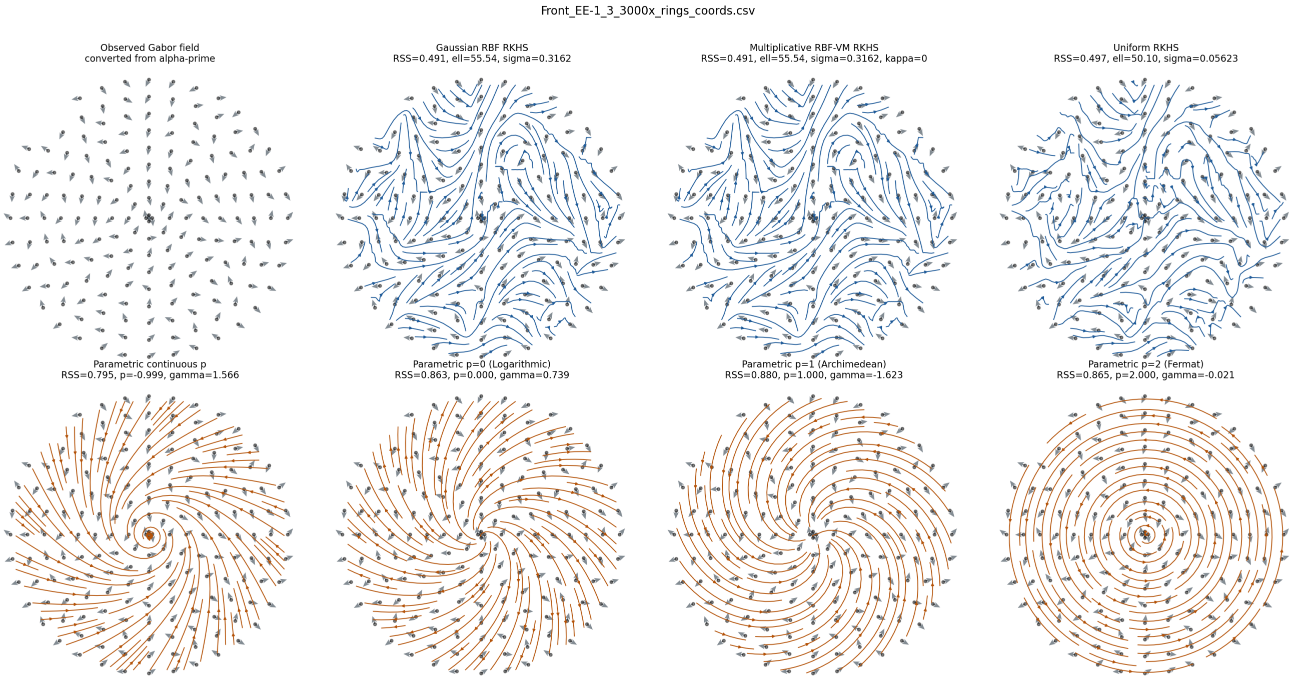

Figure 5: Fitted overlays for `Front_EE-1_3_3000x_rings_coords.csv`. The observed Gabor field is shown with arrow opacity $\alpha = 0.7$; fitted models are shown as streamlines.

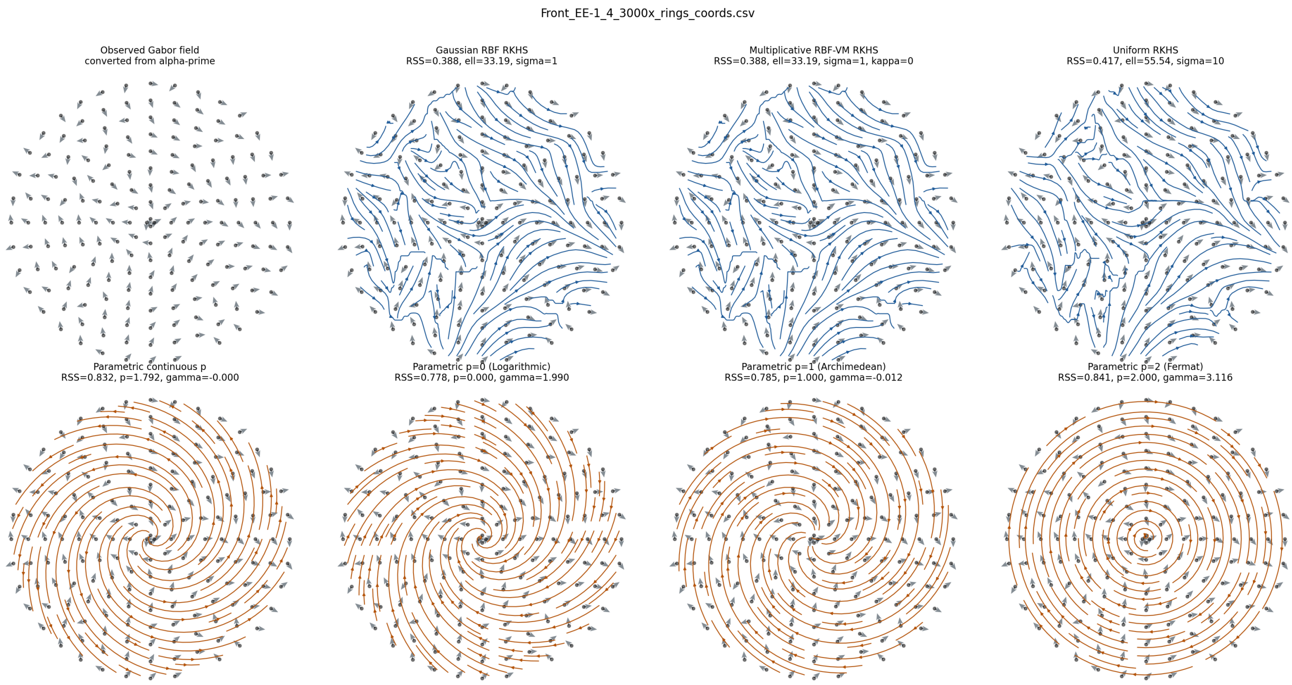

Figure 6: Fitted overlays for `Front_EE-1_4_3000x_rings_coords.csv`. The observed Gabor field is shown with arrow opacity $\alpha = 0.7$; fitted models are shown as streamlines.

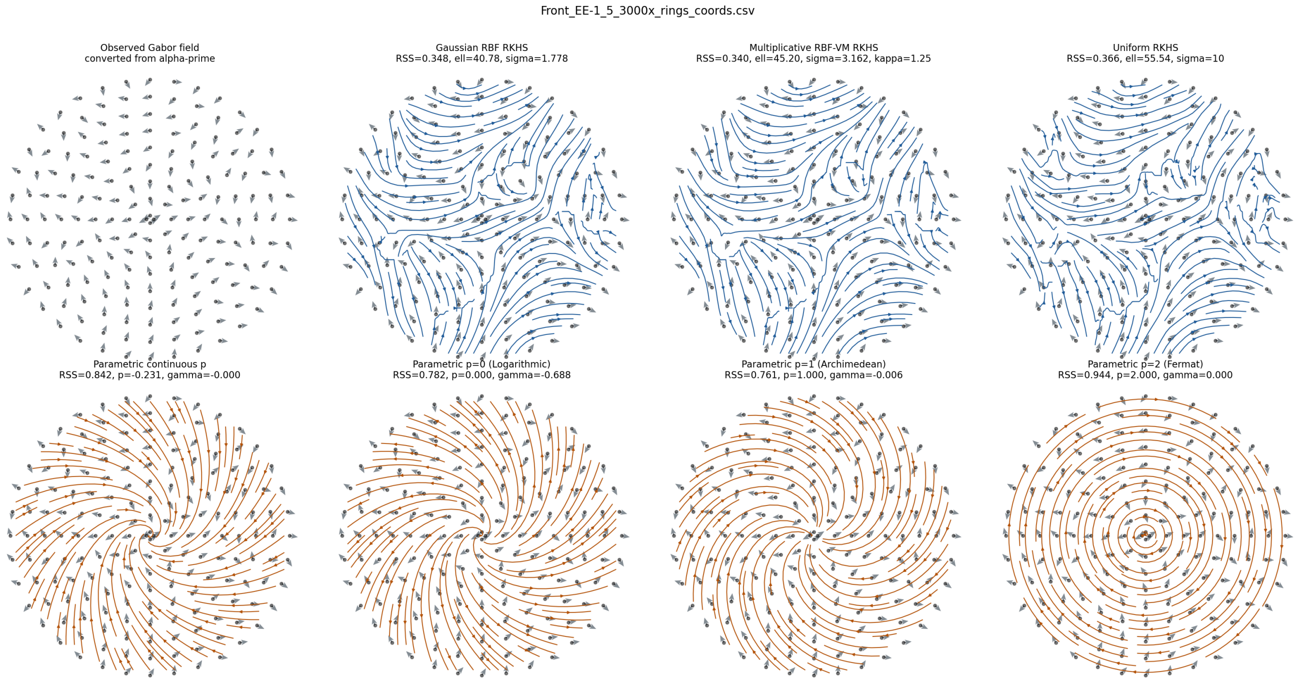

Figure 7: Fitted overlays for `Front_EE-1_5_3000x_rings_coords.csv`. The observed Gabor field is shown with arrow opacity $\alpha = 0.7$; fitted models are shown as streamlines.

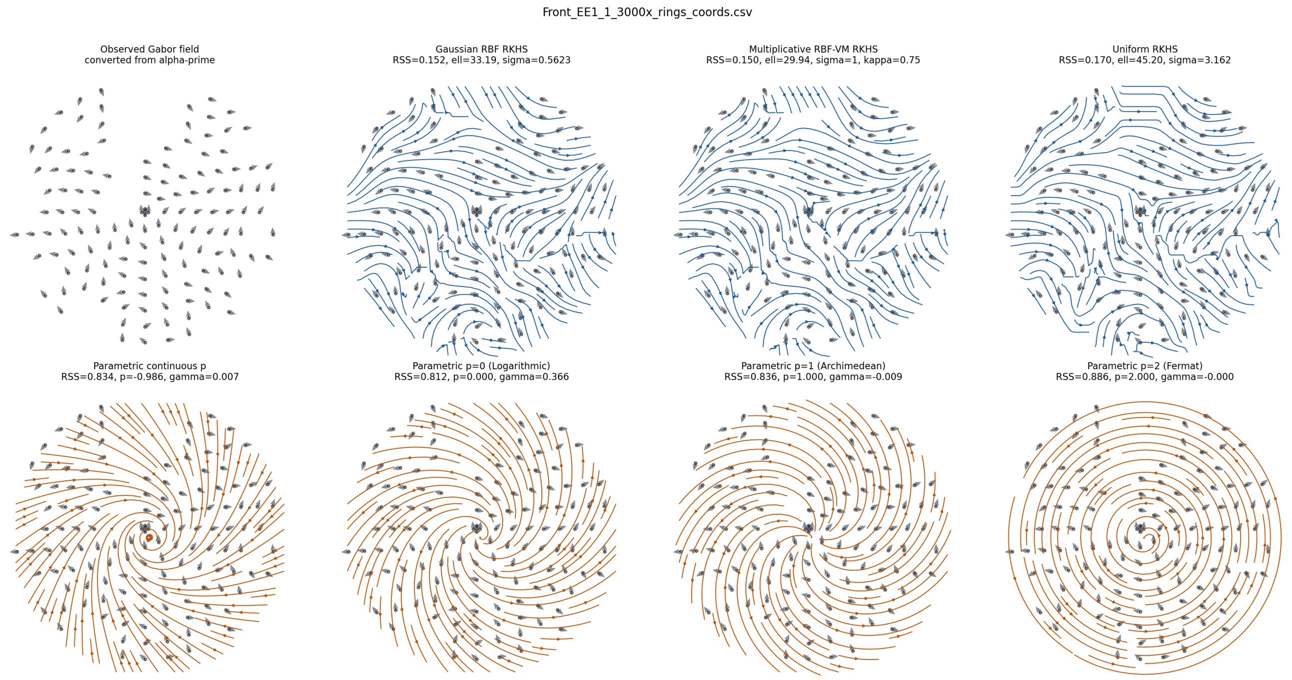

Figure 8: Fitted overlays for `Front_EE1_1_3000x_rings_coords.csv`. The observed Gabor field is shown with arrow opacity $\alpha = 0.7$; fitted models are shown as streamlines.

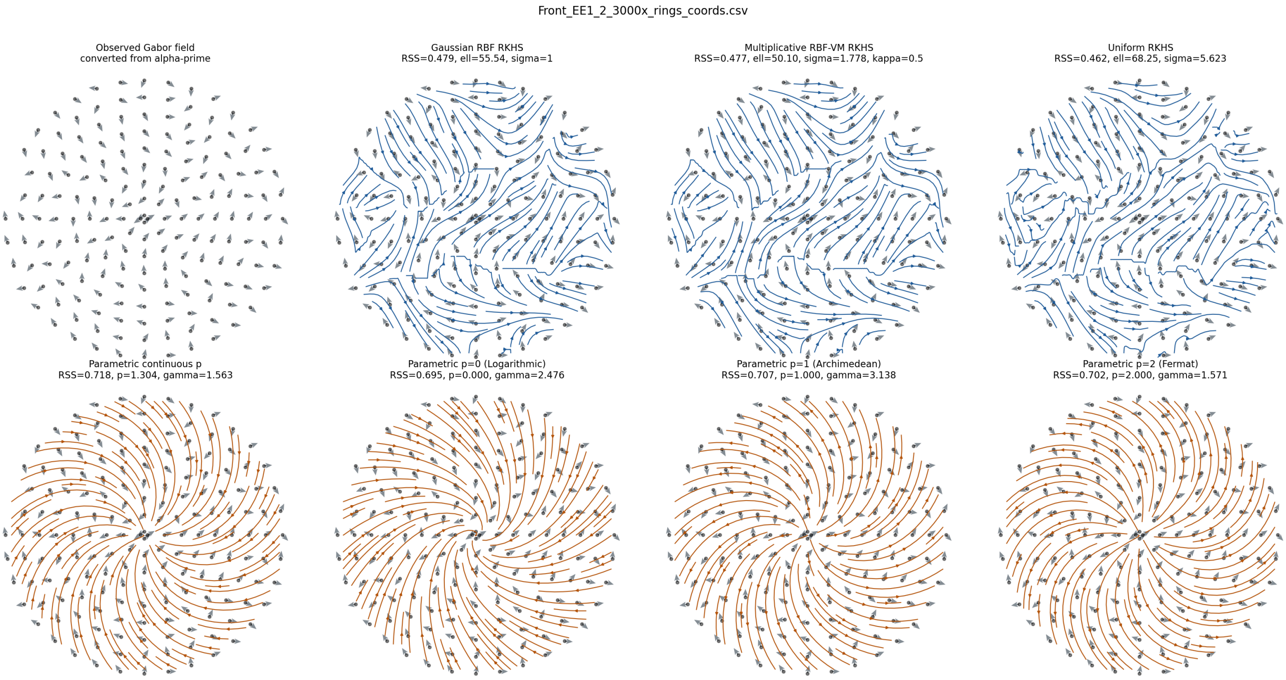

Figure 9: Fitted overlays for `Front_EE1_2_3000x_rings_coords.csv`. The observed Gabor field is shown with arrow opacity $\alpha = 0.7$; fitted models are shown as streamlines.

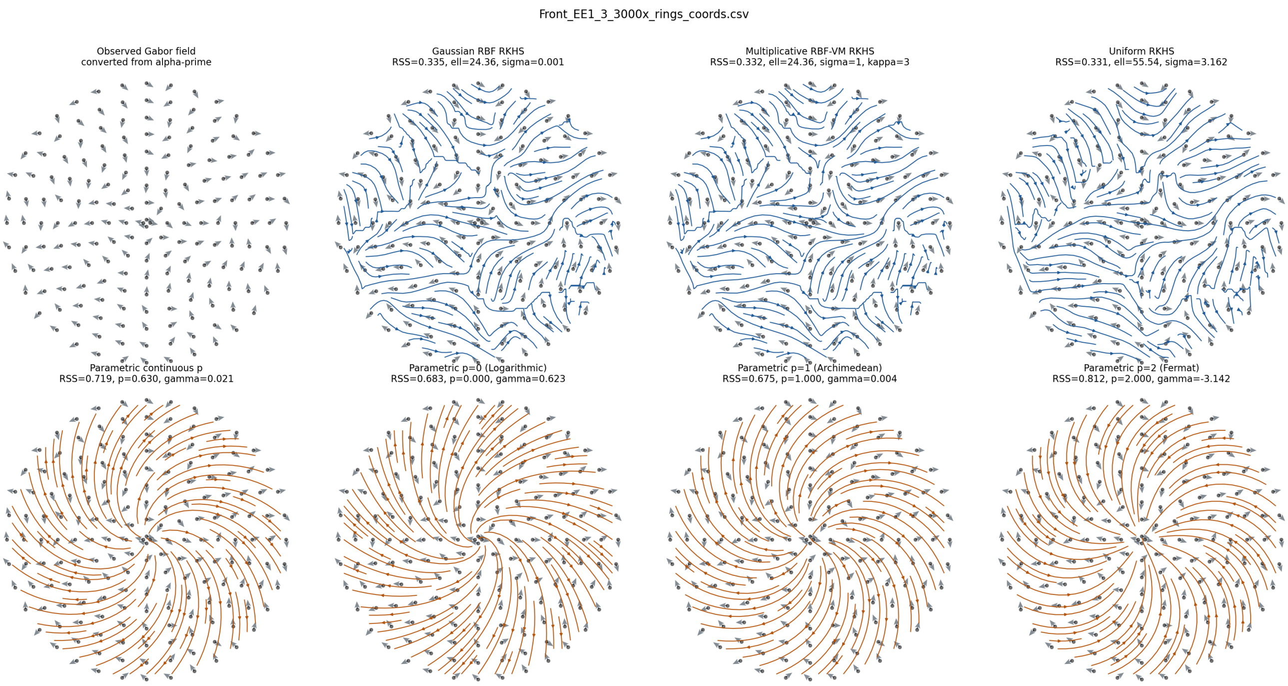

Figure 10: Fitted overlays for `Front_EE1_3_3000x_rings_coords.csv`. The observed Gabor field is shown with arrow opacity $\alpha = 0.7$; fitted models are shown as streamlines.

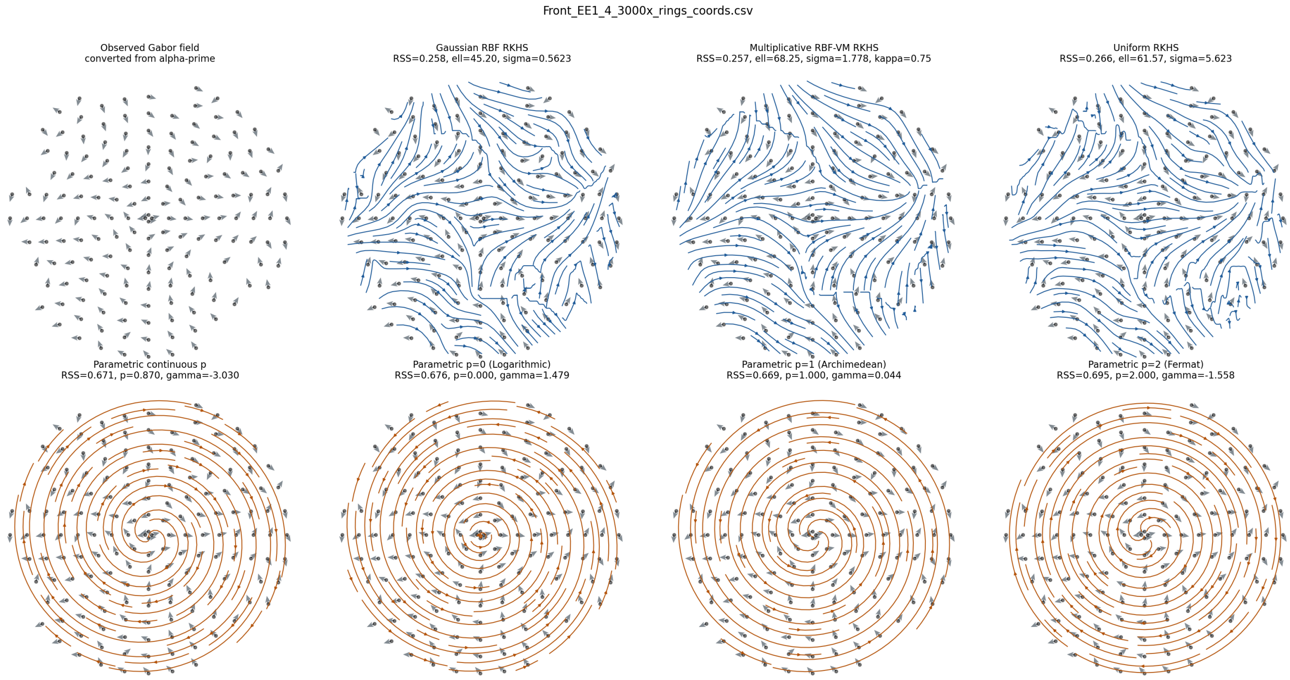

Figure 11: Fitted overlays for `Front_EE1_4_3000x_rings_coords.csv`. The observed Gabor field is shown with arrow opacity $\alpha = 0.7$; fitted models are shown as streamlines.

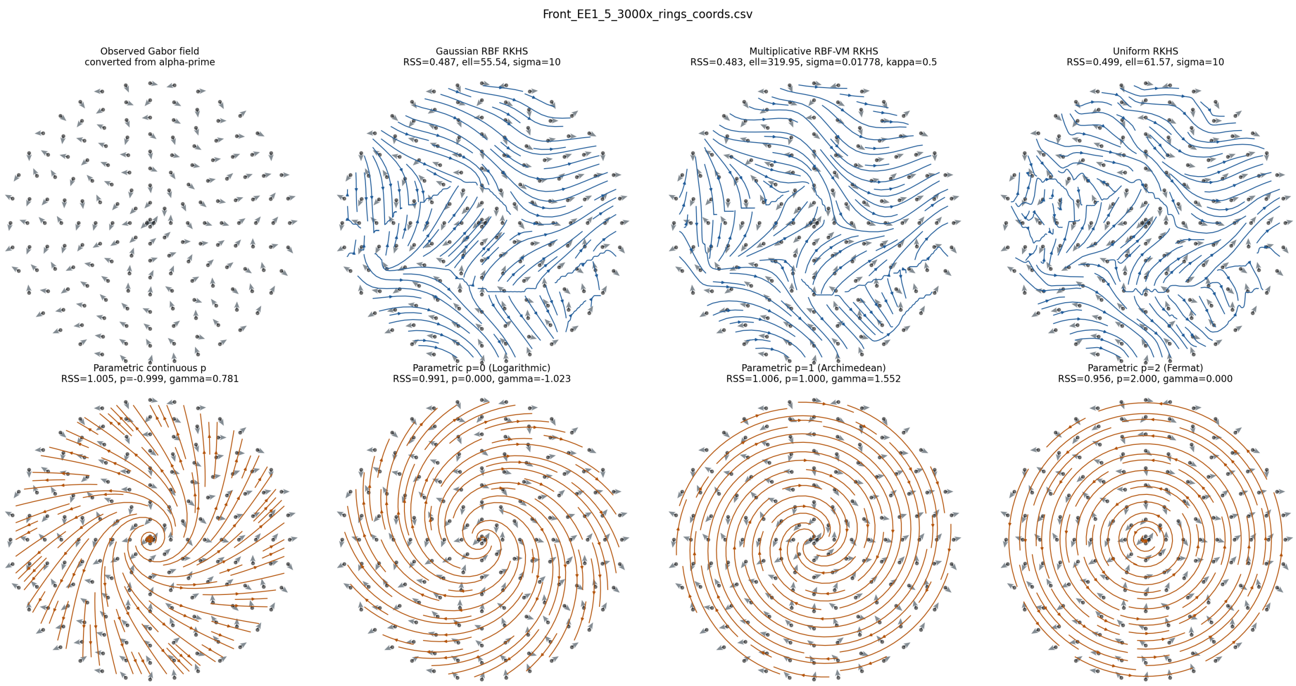

Figure 12: Fitted overlays for `Front_EE1_5_3000x_rings_coords.csv`. The observed Gabor field is shown with arrow opacity $\alpha = 0.7$; fitted models are shown as streamlines.

6 Interpretation

The fixed-grid rerun confirms the qualitative conclusion from held-out RSS: the kernel families fit the Gabor vector-field exports substantially better than the current parametric spiral families. With the wider and denser non-adaptive grid, the multiplicative RBF–von-Mises kernel has the best mean held-out RSS, with Gaussian RBF very close behind. The parametric models remain interpretable, but their held-out axial errors are roughly twice as large.